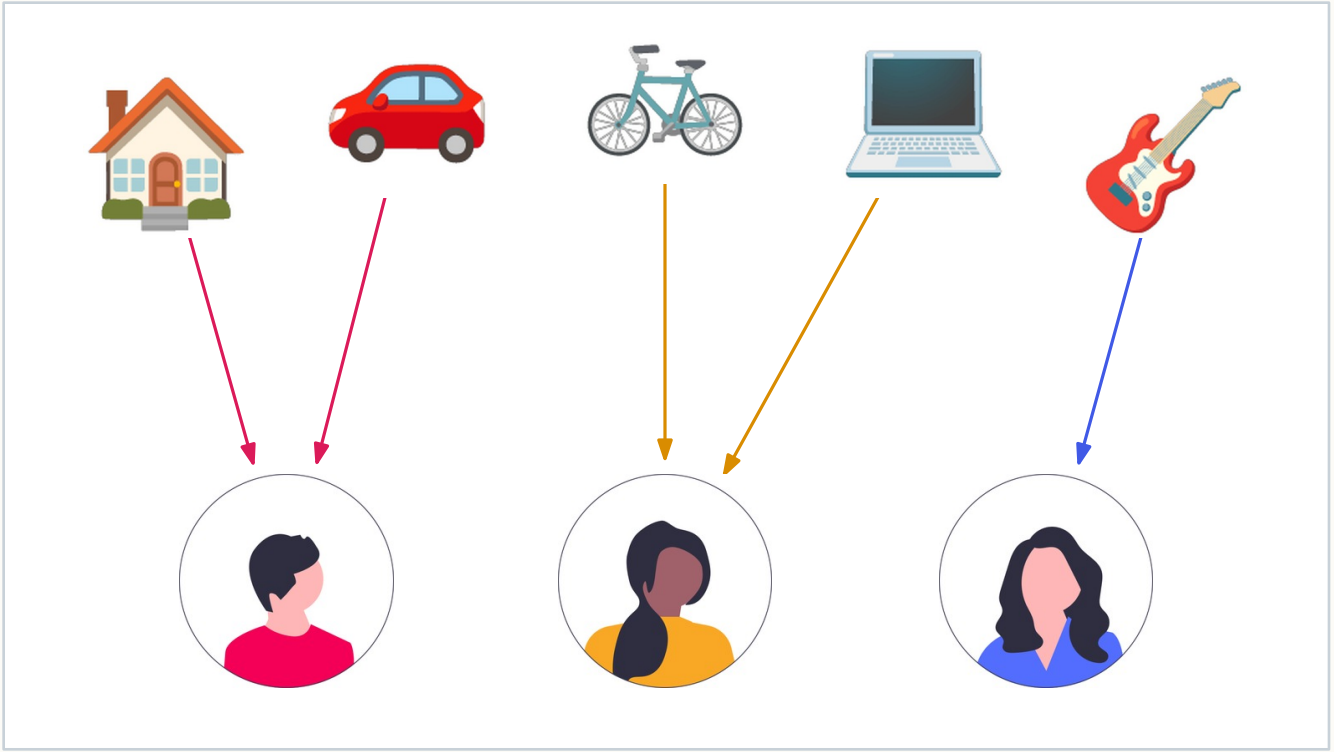
# Fair Division

Lecture 03:
Indivisible Items:
EF1 and EFX



**Hannaneh Akrami**

UNIVERSITÄT BONN

# The Big Picture

| | **divisible** | **indivisible** |
|---|---|---|
| **desirable** | cake cutting (Lectures 1–2) | **indivisible goods** (today) |
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From now on: **indivisible** items.





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Instance $I = (N, M, V)$

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- valuations $v_i : 2^M \to \mathbb{R}_{\geq 0}$

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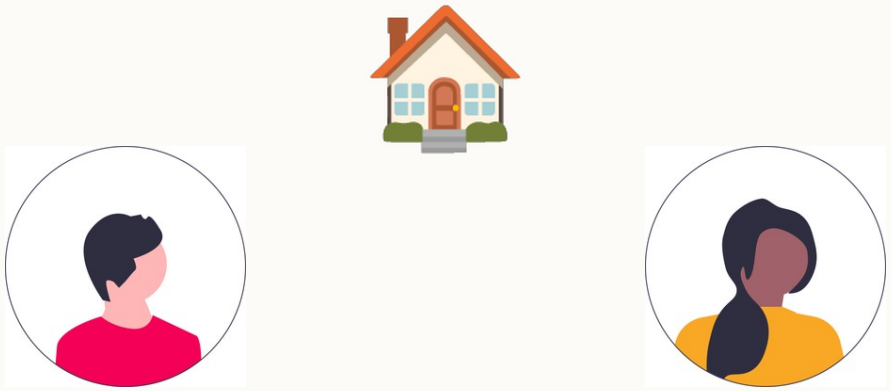
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$X_1 = \{g_1\}$, $X_2 = \{g_2, g_5\}$
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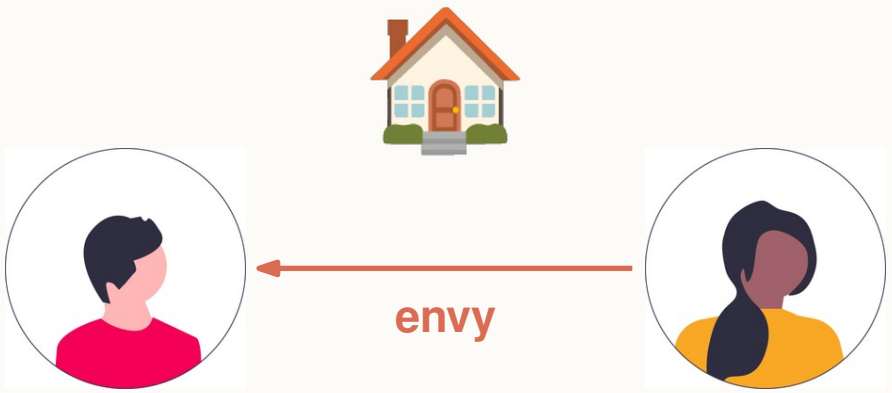
# EF and PROP Are Too Strong

**Recall.** $X$ is **envy-free (EF)** if $v_i(X_i) \geq v_i(X_j)$ for all agents $i, j$.
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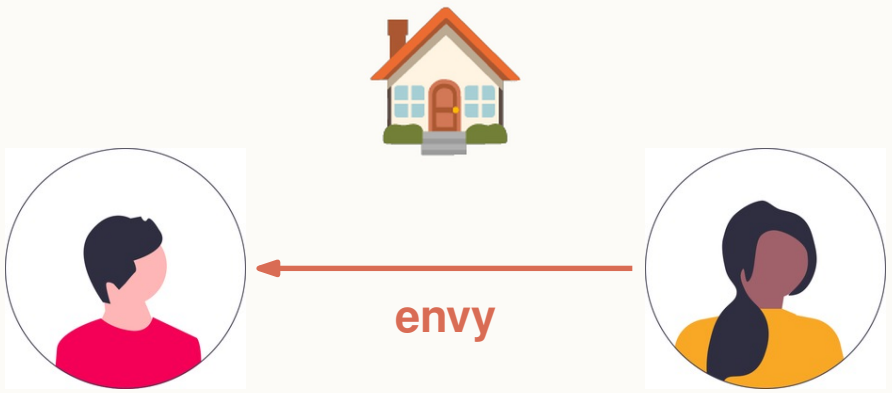
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$\Rightarrow$ **relax** envy-freeness

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EFX: even removing the **least valuable** good must kill the envy.

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**Open:** additive valuations, $n \geq 4$.

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**Algorithm (round robin).** Agents pick in turns $1, 2, \ldots, n, 1, 2, \ldots, n, \ldots$; each takes her **favourite** remaining good.

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= our EF1 example:
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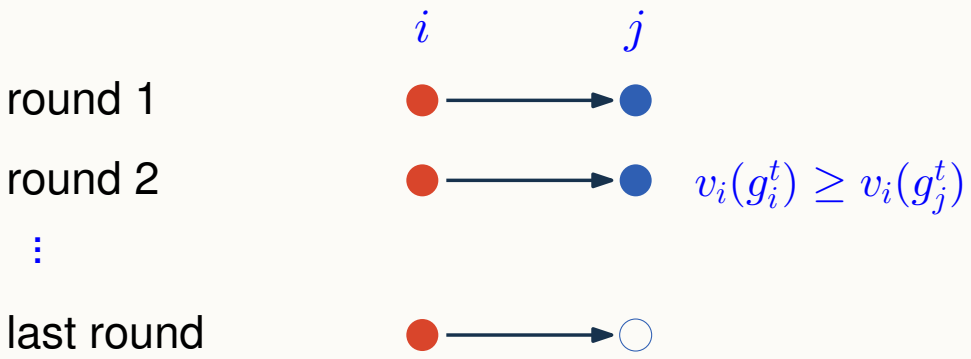

**Question.** Does $i$ envy $j$?

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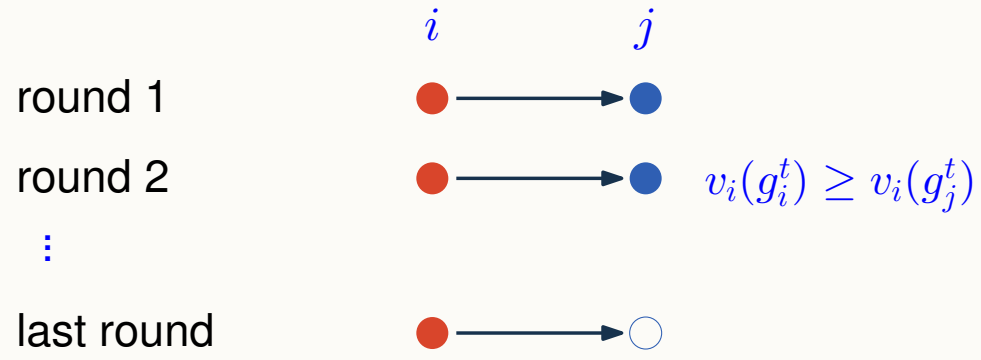
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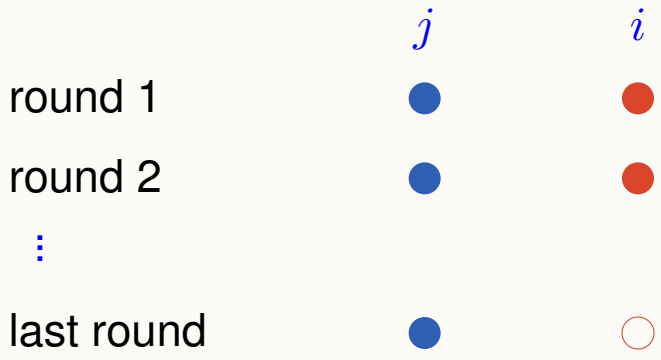
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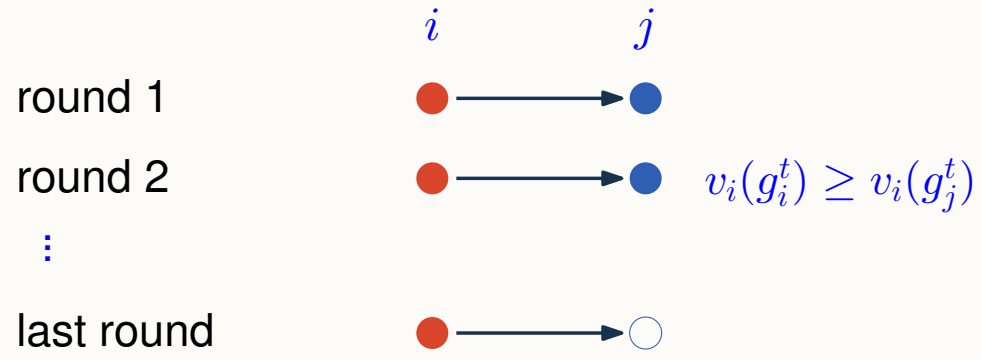
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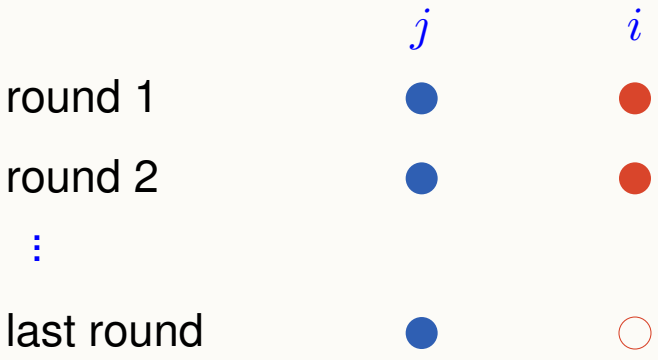
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**Case 2: $j$ before $i$**



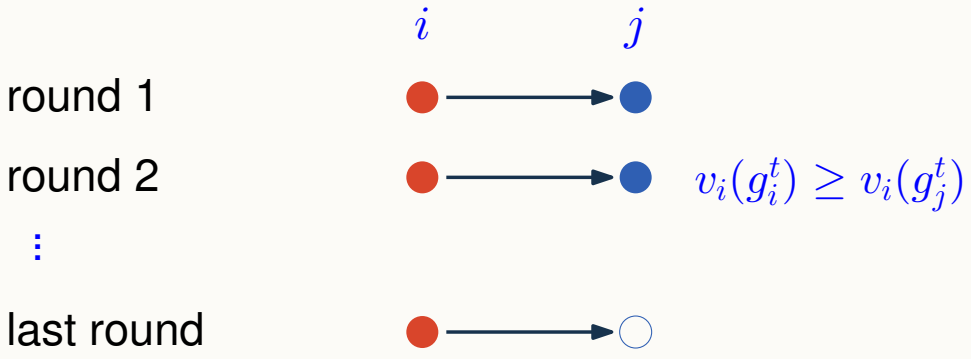

**Question.** Which good of $j$ to remove?

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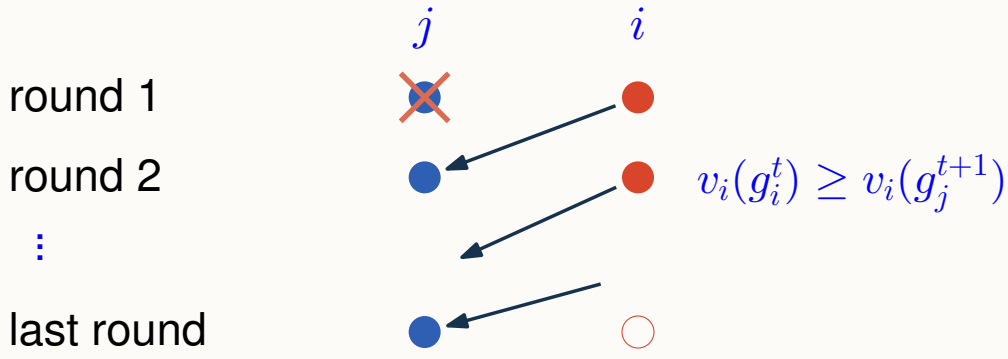
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**Case 2: $j$ before $i$**




$v_i(g_i^t) \geq v_i(g_j^{t+1})$

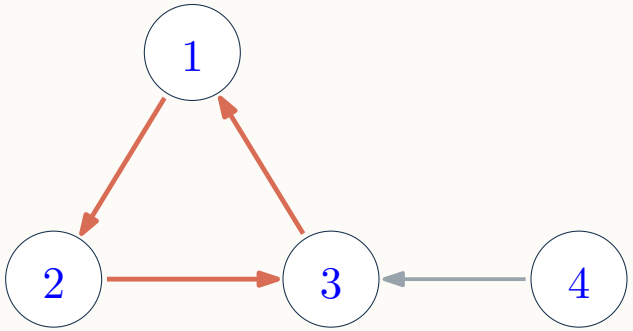
**First pick $g_j^1$:**
$v_i(X_i) \geq \sum_t v_i(g_j^{t+1}) = v_i(X_j \setminus \{g_j^1\})$

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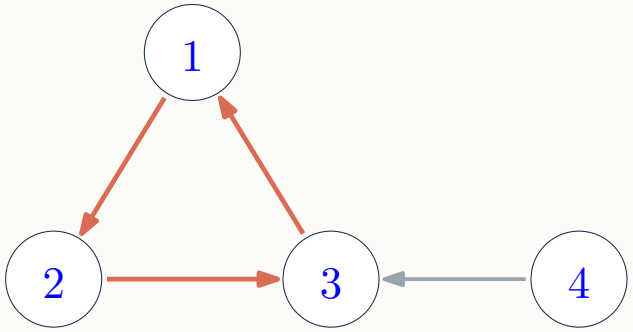
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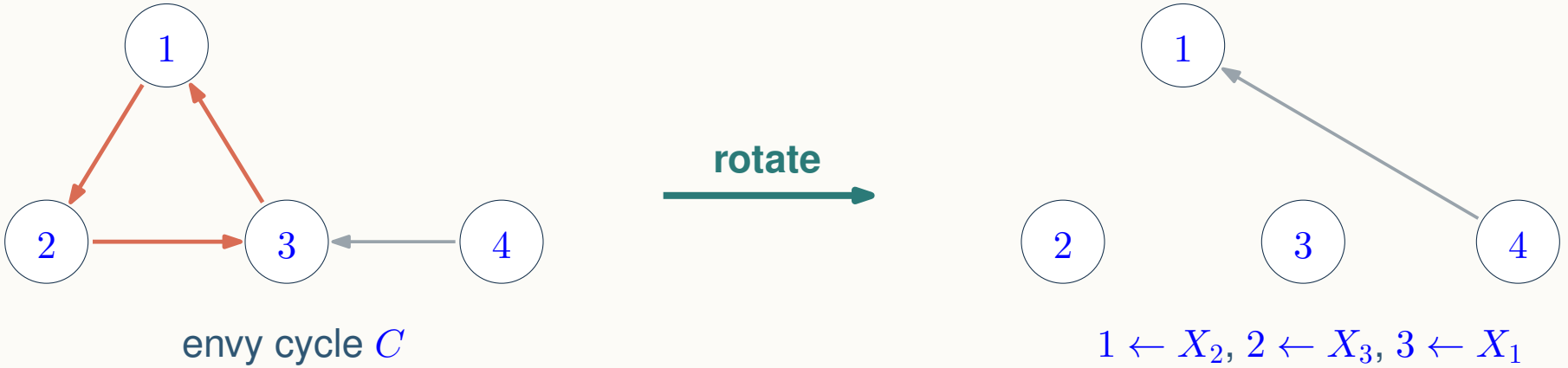



envy cycle $C$

**Question.** How to use a cycle?

# The Envy Graph

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envy cycle $C$

$1 \leftarrow X_2$, $2 \leftarrow X_3$, $3 \leftarrow X_1$

**Rotate**: everyone on $C$ takes the bundle she envies $\to$ $X^C$

# Envy-cycle Elimination

**Algorithm (envy-cycle elimination)** (Lipton, Markakis, Mossel, Saberi 2004). Start with the empty allocation. For each good $g$:

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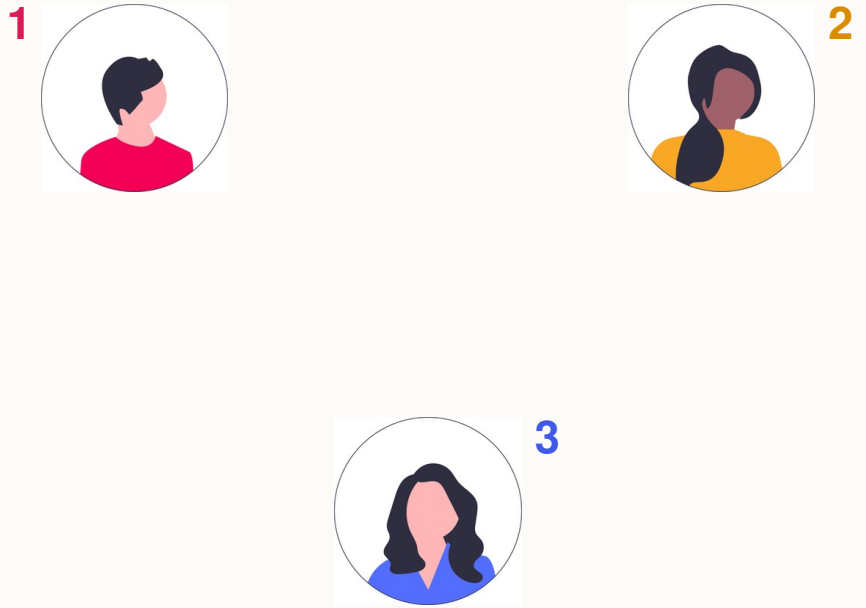
**Does step 1 stop?** Each rotation **removes edges**.

**Why a source?** Nobody envies $s$: remove $g$ again.

# Envy-cycle Elimination: Example

Order $g_3, g_1, g_2, g_4, g_5$

| | $g_1$ | $g_2$ | $g_3$ | $g_4$ | $g_5$ |
|---|---|---|---|---|---|
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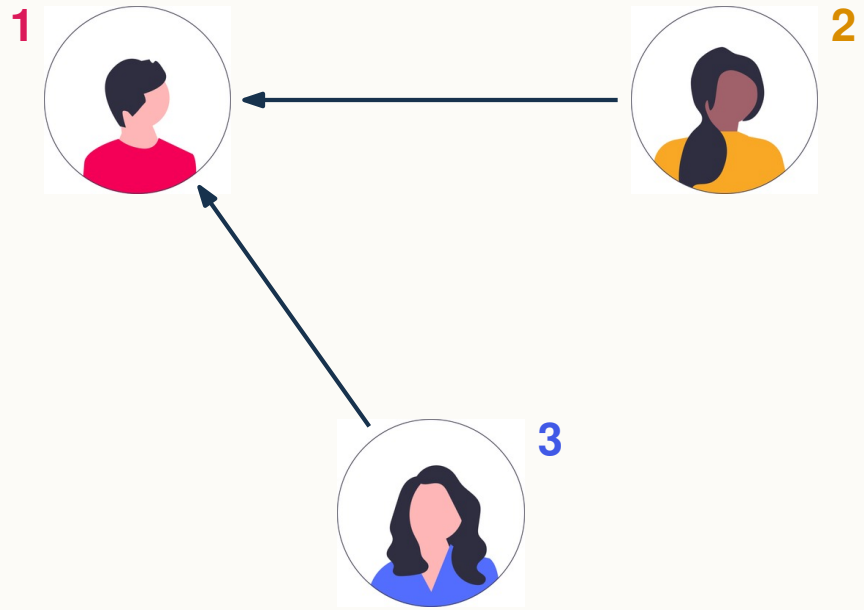



Start: empty

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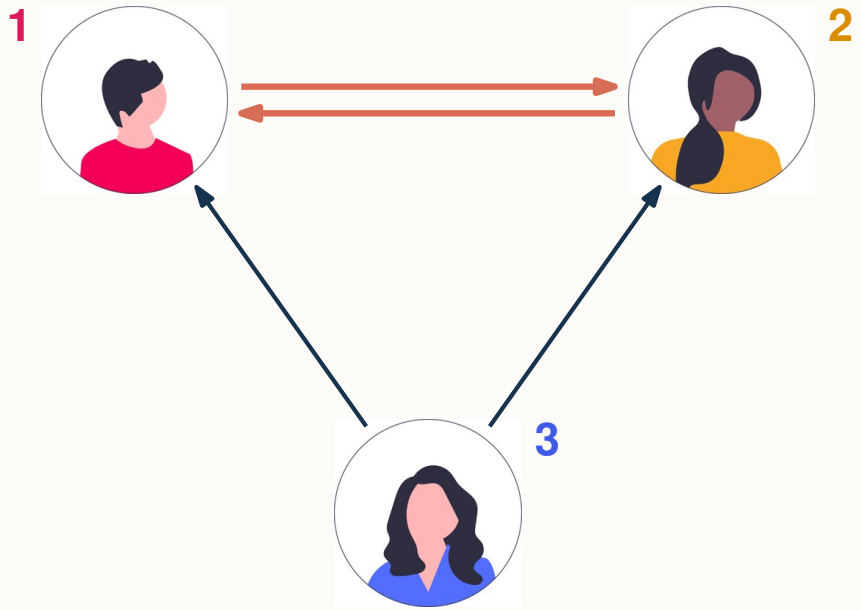



$g_3 \to 1$

# Envy-cycle Elimination: Example

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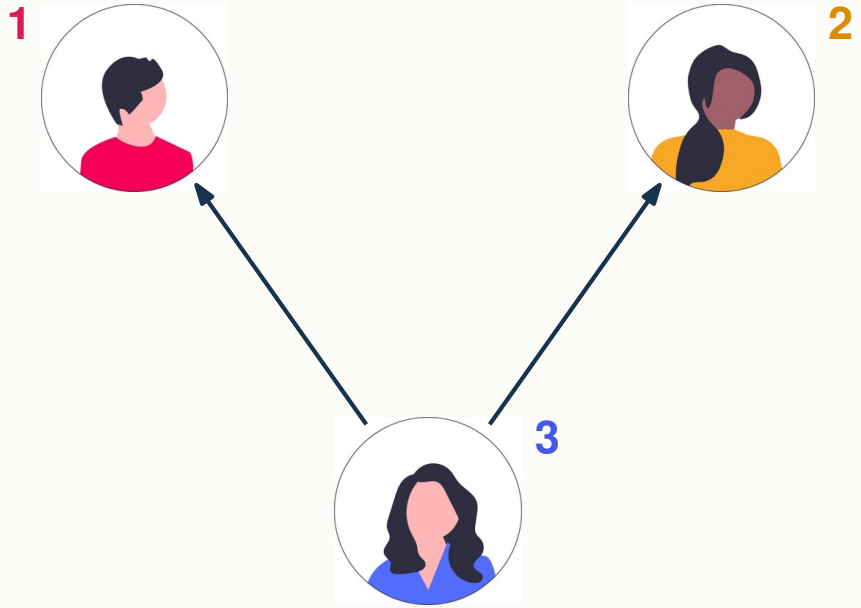



$g_1 \to 2$: **cycle!**

# Envy-cycle Elimination: Example

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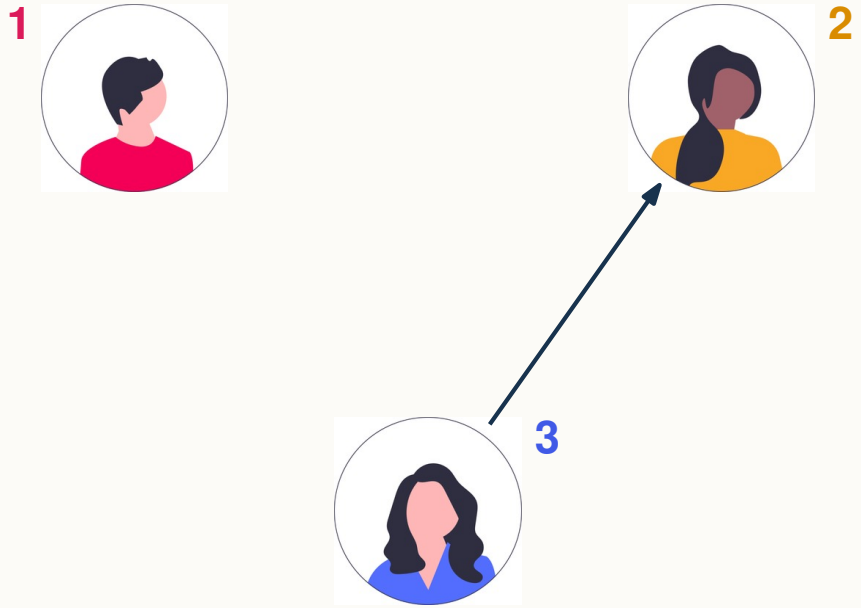



## Rotate

# Envy-cycle Elimination: Example

Order $g_3, g_1, g_2, g_4, g_5$

| | $g_1$ | $g_2$ | $g_3$ | $g_4$ | $g_5$ |
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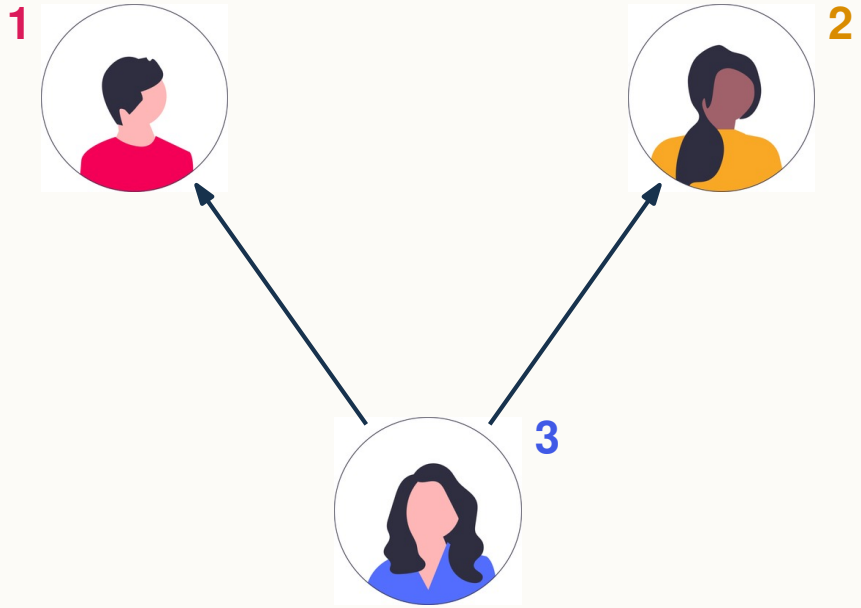



$g_2 \to 3$

# Envy-cycle Elimination: Example

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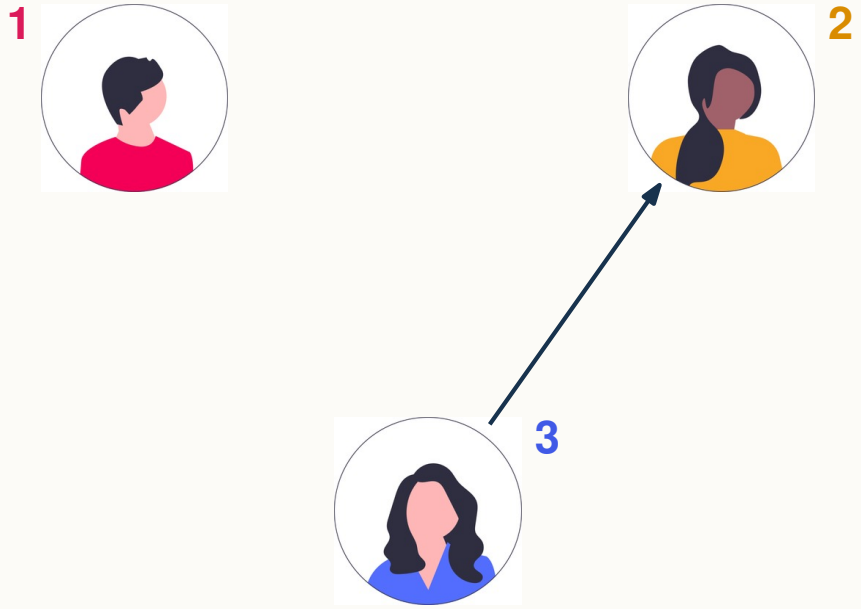



$g_4 \to 1$

# Envy-cycle Elimination: Example

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$g_5 \to 3$: **EF1** ✓

# Rotating an Envy Cycle

**Lemma.**
Let $C$ be a cycle in $G_X$. Then
(a) every agent weakly prefers her bundle in $X^C$; agents on $C$ strictly prefer it;
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**(c)** Off $C$: no change. On $C$: value $\uparrow \Rightarrow$ no new envy,
and no envy for the bundle she now holds.

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Let $X$ be EF1, $s$ a source of $G_X$ and $g$ unallocated. Giving $g$ to $s$ keeps $X$ EF1.

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**Question.** Where did we use additivity?

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**Nowhere!** Only monotonicity.

# Goods and Chores

**Definition (doubly monotone)** (Bhaskar, Sricharan, Vaish 2021). $v_i : 2^M \to \mathbb{R}$ is **doubly monotone** if $M = G_i \uplus C_i$ such that for all $S \subseteq M$ and $o \notin S$:

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Remove a **good** from $X_j$, or a **chore** from $X_i$.

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**Is rotating any cycle safe?**
**No!** The EF1 witness (a chore) can get lost.

# Rotating Any Cycle Can Break EF1

| | $c_1$ | $c_2$ | $c_3$ | $c_4$ |
|---|---|---|---|---|
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| 2  | $-1$ | $-1$ | $-5$ | $-1$ |
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**EF1 ✓**

- 1: remove $c_1$
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**Question.** Still EF1?

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no single removal helps

Agent 1's favourite bundle is $X_3$, not $X_2$: **not top-trading**.

# The Top-trading Envy Graph

**Definition (top-trading envy graph).** $T_X$ has an edge $i \to j$ if $X_j$ is a **most preferred** bundle of $i$ and $i$ envies $j$:

$$v_i(X_j) = \max_{k \in N} v_i(X_k) > v_i(X_i).$$

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**Yes**: every out-degree in $T_X$ is $\geq 1$.

# Top-trading Envy-cycle Elimination

**Objective chore**: chore for all. Else **goods-type**: $N_o = \{i : o \in G_i\}$.

**Phase 1** (Bhaskar, Sricharan, Vaish 2021). For each goods-type item $o$:
(a) while $G_X$ has a cycle, rotate along it;
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**Phase 2**. For each objective chore $c$:
(a) if $G_X$ has no sink, rotate a cycle of $T_X$;
(b) give $c$ to a **sink** of $G_X$.

# Analysis of Phase 1

$$(*) \quad v_i(X_i) \geq v_i(X_j) \text{ or } \exists\, g \in X_j :\ v_i(X_i) \geq v_i(X_j \setminus \{g\})$$

**Lemma.**
Rotations and step (b) preserve $(*)$. In step (b), a source of $G_X[N_o]$ exists.

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- $i \in N_o$: no envy towards $s \Rightarrow$ remove $o$.
- $i \notin N_o$: $v_i(X_i) \geq v_i(X_s \setminus \{g\}) \geq v_i\big((X_s \setminus \{g\}) \cup \{o\}\big)$

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$$(*) \quad v_i(X_i) \geq v_i(X_j) \text{ or } \exists\, g \in X_j : \; v_i(X_i) \geq v_i(X_j \setminus \{g\})$$

**Lemma.**
Rotations and step (b) preserve $(*)$. In step (b), a source of $G_X[N_o]$ exists.

**Rotation:** as before.

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**Question.** Why does a source exist?

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**Existence:** $G_X[N_o]$ is acyclic and non-empty.

# Analysis of Phase 2

**Lemma.**
(a) If $G_X$ has no sink, $T_X$ has a cycle; after rotating it, all agents on it are sinks.
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**Theorem** (Bhaskar, Sricharan, Vaish 2021).
For doubly monotone valuations, top-trading envy-cycle elimination computes a complete EF1 allocation in polynomial time.

# Summary

| Algorithm | Valuations | Guarantee |
|---|---|---|
| Round robin | additive goods | EF1 |
| Envy-cycle elimination | monotone goods | EF1 |
| Top-trading envy-cycle elim. | doubly monotone | EF1 |

# Exercises

**Question 1** (EFX for identical orderings). Suppose all agents have additive valuations and the **same preference order** over the goods: there is an ordering $g_1, g_2, \ldots, g_m$ of $M$ such that $v_i(g_1) \geq v_i(g_2) \geq \cdots \geq v_i(g_m)$ for every agent $i$. Prove that a complete EFX allocation exists and can be computed in polynomial time.

**Question 2** (EFX for two agents). Prove that for **two agents** with additive valuations, a complete EFX allocation always exists and can be computed in polynomial time.